

Credit Card & Customer Analytics

End-to-End Data Engineering & Business Intelligence Project Report

1. Project Overview & Operational Objective

The objective of this project was to establish a fully integrated end-to-end data analytics pipeline capable of transforming raw transactional records into actionable business intelligence. By establishing automated extraction, cleaning, and structured injection procedures into an enterprise relational database, this infrastructure addresses critical requirements to monitor multi-million dollar asset portfolios.

The downstream analytical models built within Power BI provide management with continuous, robust monitoring over primary financial key performance indicators (KPIs), week-over-week growth metrics, high-value demographic behavioral trends, and early-warning asset health triggers to optimize global operational risk strategies.

2. System Architecture & Technical Stack

The architecture follows a standard data engineering layout optimized for performance, scalability, and seamless metric delivery:

- **Data Pipeline Sources:** Structured enterprise comma-separated value streams parsing transaction logs (`credit_card.csv`) and demographic client files (`customer.csv`).
- **Relational Database Engine (SQL):** Target system managed under a MySQL server database deployment layer responsible for constraint execution, primary/foreign key mappings, indexing, and internal ingest storage.
- **Business Intelligence Engine (Power BI):** Power BI Desktop pipeline handling physical database connection queries, multi-source dimensional data blending inside Power Query, star-schema optimization, DAX calculation processing, and dynamic visual layouts.

3. Data Engineering & Relational Database Design

A target relational database environment named `ccdb` was deployed using physical structure layouts specifically designed to process millions of rows. The schema utilizes a 1-to-Many relational join mapping linked explicitly via the unique client identifier field `Client_Num`.

SQL Target Schema Definition Scripts

```
CREATE DATABASE ccdb;
USE ccdb;

-- 1. Credit Card Transaction Table Definition
CREATE TABLE cc_detail (
    Client_Num INT,
    Card_Category VARCHAR(20),
    Annual_Fees INT,
```

```

    Activation_30_Days INT,
    Customer_Acq_Cost INT,
    Week_Start_Date DATE,
    Week_Num VARCHAR(20),
    Qtr VARCHAR(10),
    current_year INT,
    Credit_Limit DECIMAL(10,2),
    Total_Revolving_Bal INT,
    Total_Trans_Amt INT,
    Total_Trans_Ct INT,
    Avg_Utilization_Ratio DECIMAL(10,3),
    Use_Chip VARCHAR(10),
    Exp_Type VARCHAR(50),
    Interest_Earned DECIMAL(10,3),
    Delinquent_Acc VARCHAR(5)
);

-- 2. Customer Demographics Table Definition
CREATE TABLE cust_detail (
    Client_Num INT,
    Customer_Age INT,
    Gender VARCHAR(5),
    Dependent_Count INT,
    Education_Level VARCHAR(50),
    Marital_Status VARCHAR(20),
    State_cd VARCHAR(50),
    Zipcode VARCHAR(20),
    Car_Owner VARCHAR(5),
    House_Owner VARCHAR(5),
    Personal_Loan VARCHAR(5),
    Contact VARCHAR(50),
    Customer_Job VARCHAR(50),
    Income INT,
    Cust_Satisfaction_Score INT
);

```

High-Velocity Data Ingestion Workflow: Server workloads pull active csv logs directly into the tables using optimized `LOAD DATA LOCAL INFILE` server commands. To preserve formatting, raw variable text streams are dynamically intercepted and parsed through the SQL `STR_TO_DATE(@Week_Start_Date, '%d-%m-%Y')` functions, converting them into native SQL chronological database structures.

4. Portfolio Performance Dashboard & YTD KPI Summary

The analytical platform processes operational transaction data across all active client categories, consolidating granular log records into high-level dashboard metrics for real-time portfolio management.

Primary Financial Key Performance Indicators (YTD Overview)

\$57M OVERALL REVENUE	\$46M TRANSACTION AMT	\$8M TOTAL INTEREST	667.2K TRANSACTION COUNT
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Portfolio Health & Growth Growth Velocity Triggers

+28.8% Week-over-Week (WoW) Growth	57.5% Overall Activation Rate	6.06% Portfolio Delinquency Rate
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5. Visual Dashboard Analysis & Multi-Dimensional Deep-Dive

A. Transactional Characteristics & Channel Volume Splits

- **Tier Performance Dominance:** High-volume, entry-to-mid consumer portfolios drive the vast majority of consumer transaction velocity. **Blue and Silver** product categories collectively command **93% of overall transactions**. The Blue tier alone secures **\$47M** in gross revenue, with Silver contributing **\$6M**.
- **Payment Channel Utilization:** Physical point-of-sale card presence via **Swipe transactions** remains heavily preferred, accounting for **\$36M** of portfolio volume. **Chip insertion** follows at **\$17M**, while digital **Online channels** lag significantly at **\$4M**.
- **Expense Distribution:** Primary spending sub-verticals lean heavily toward standard baseline utilities and consumer discretionary items, led by recurring **Bills (\$14M)**, followed closely by **Entertainment (\$10M)** and **Fuel (\$10M)**.

B. Customer Demographic Clusters & Geographic Performance Density

- **Gender Contribution Breakdown:** Portfolio spend distribution metrics indicate that **Male clients** generate higher total volume at **\$31M**, while **Female clients** contribute **\$26M** to total portfolio revenue.
- **Geographic Concentration:** Macroeconomic geographical performance indicates a significant concentration of volume, with three major commercial states—**Texas (TX)**, **New York (NY)**, and **California (CA)**—generating **68% of overall revenue**.
- **Age Bracket Distributions:** The core demographic sweet spot centers inside mature career stages. The **40-50 age bracket** represents the primary engine, generating **\$25M**, closely followed by the **50-60 age group** at **\$19M**.

- **Socioeconomic Profiles:** Professionals categorized as **Business owners** generate the largest professional revenue share at **\$18M**, with standard **White-collar professionals** capturing **\$10M**.

6. Actionable Strategic Data-Driven Recommendations

1. Accelerate Digital & Online Channel Ingestion Frameworks

With digital Online transaction volume sitting at an underperforming \$4M compared to physical Swipe volume (\$36M), the business is exposed to high acquisition friction. The marketing team should launch virtual card offerings and targeted e-commerce rewards programs to boost digital card usage.

2. Execute High-Margin Premium Tier Migration Campaigns

Because 93% of transaction volume is concentrated in the basic Blue and Silver cards, significant upselling potential exists. Management should run automated database queries to identify high-income individuals within these tiers and offer them seamless migrations to premium Gold and Platinum cards.

3. Optimize Regional Marketing Budgets

Because TX, NY, and CA contribute 68% of overall revenue, marketing spend should be focused primarily on these states. Cultivating regional merchant networks, co-branded partnerships, and geo-targeted rewards will maximize portfolio acquisition returns.

4. Deploy Proactive Credit Risk and Delinquency Mitigation Early-Warning Models

With the portfolio delinquency rate at 6.06%, risk management teams need to establish proactive algorithmic guards. Implementing predictive models that flag accounts based on rapid spikes in utilization ratio will allow for automated limit reductions before accounts slip into costly collection cycles.